

SHASHIDHARA NARAYANAPPA

Lead Software Engineer | Full-Stack React, TypeScript, and Python

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PROFESSIONAL SUMMARY

Lead-level full-stack engineer with 5+ years delivering production web applications in React, TypeScript, Python, and C#. Provided day-to-day technical direction to a seven-engineer Agile team while remaining hands-on in the codebase, mentoring through pairing and code review and establishing the design system, component library, coding standards, and code review discipline the team built to. Owns delivery end to end, from requirements and technical design through implementation, automated testing, deployment, and production support. Led legacy modernization work that cut page load times by roughly half, and builds Python backends with FastAPI, SQLAlchemy, and Celery against relational databases. M.S. in Human-Computer Interaction, with peer-reviewed research in educational technology for children.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C#, SQL, PHP, HTML5, CSS3

Frontend: React.js, Next.js, React Hooks(App Router, SSR/SSG, React Server Components), Redux, Angular, Tailwind, Component Libraries, Shared Design Systems, Componentization, Responsive and Mobile-First Development, Accessibility (WCAG 2.1 AA, ADA)

Backend: Python (FastAPI, SQLAlchemy, Celery, pandas, NumPy), Node.js, REST API Design, GraphQL, Asynchronous and Background Job Processing, Server-Sent Events, WebSockets, OAuth, JWT

Data: PostgreSQL, MySQL, Firebase, pgvector, SQL, Relational Schema Design, Data Modeling, Query Optimization, Database Architecture Review

Cloud and DevOps: AWS (Lambda), GCP, Vercel, Netlify, Podman, Git, CI/CD Pipelines, Automated Testing, Jest, Mocha, Chai, Code Coverage, ESLint and Static Analysis, Code Review Discipline

Engineering Practice: SDLC, Agile/Scrum, Waterfall, Sprint Planning and Estimation, Technical Specifications and Design

Documentation, Application Architecture, Legacy Modernization, Mentoring and Pairing, Production Support and Incident Triage

AI and LLM: Claude Code, Claude API, Cursor, GitHub Copilot, MCP, Ollama, LangGraph, LLM Integration, Eval Harness Engineering

3D and Immersive: Unity, C#, Unreal Pixel Streaming, Three.js, Babylon.js, WebGL

PROFESSIONAL EXPERIENCE

Indiana University Indianapolis, SETH Lab | Software Developer and UX Researcher

Aug 2024 - May 2026

- Built and maintained the lab public web application end to end on a component-based frontend architecture, gathering requirements from faculty stakeholders, translating them into designs, and owning production maintenance; raised the accessibility score from 70 to 95 through ADA and WCAG remediation.
- Built Python data pipelines processing frame-level signal data with rule-based classification, and a Unity application in C# with behavioral event logging and local data persistence for post-session analysis.
- Taught Unity development as a course instructor at IU Indianapolis, mentoring students through build, debugging, and deployment workflows.
- Handled sensitive human-subjects data under IRB protocol, including consent flow, restricted persistence, and de-identified export, applying formal information handling and retention procedures.
- Ran usability testing and stakeholder acceptance reviews before release, then triaged and resolved defects found post-launch.

Zero Distance | Lead Frontend Developer

Feb 2022 - Jul 2024

- Provided day-to-day technical direction to a seven-engineer Agile team delivering 8+ production applications across concurrent product teams, mentoring through pairing, code review, and design guidance while remaining hands-on in the codebase.
- Established the engineering standards the team built to: a shared React and Tailwind CSS design system, a reusable component library, ESLint configuration, componentization patterns, and code review discipline adopted across all products.
- Led technical design for new applications and wrote and reviewed technical specifications and integration documentation for system components; made framework, pattern, and tooling decisions across the frontend stack.
- Owned end-to-end delivery of complex user-facing features from requirements and design through implementation, testing, deployment, and production support; contributed to sprint planning, backlog refinement, and estimation.
- Built a full-stack enterprise digital twin application for Tata Steel in React and Next.js with real-time machine maintenance status, personnel safe and unsafe zone occupancy, and danger-zone safety alerts.
- Designed and built an Events CMS in React, Node.js, and MySQL supporting 2 live events with 1,000+ attendees, designing the relational schema and query logic and contributing to CI/CD pipelines for automated testing and cloud deployment.
- Built a backend CMS in Unity and C# for an on-premise system deployed at a global steel manufacturer; authored the operator documentation and served as on-call support for incident triage and post-deployment defect resolution.

- Designed and built a cloud voice agent on AWS Lambda with voice-to-text, routing spoken queries to language models and returning organization-level summary answers to executives.
- Built a real-time multiplayer 3D web application with WebSocket state synchronization and custom interface components, serving 100+ concurrent users.
- Led client-facing requirement gathering with business stakeholders and executives, translating operational needs into functional specifications and production applications.

Zero Distance | Software Developer

July 2020 - Feb 2022

- Led modernization of legacy WordPress applications onto React single-page applications, cutting page load times by approximately 50% and reducing accumulated technical debt.
- Built a reusable React and Firebase component for live messaging, notifications, read state management, and presence, contributed to the shared library and adopted across multiple products by 1,000+ users.
- Led database architecture reviews to restructure read and write patterns and real-time data models, resolving performance and cost defects while maintaining data consistency.
- Introduced and documented coding standards, componentization patterns, and ESLint rules, Redux state management, version control adopted organisation wide; promoted to Lead.

PROJECTS

Event Intelligence Agent | Python Services with Automated Quality Gates

2026

- Built async Python services (FastAPI, SQLAlchemy, Celery) against Postgres, spanning 21 REST endpoints and a six-table relational schema, with an Angular 22 operator interface.
- Evaluated and applied emerging tooling to the problem, architecting a multi-stage processing pipeline on LangGraph with vector search over pgvector for retrieval.
- Built an automated regression suite over 30 hand-labeled fixtures that caught and reverted an accuracy regression before release, holding 30/30 accuracy across two migrations.
- Enforced pre-merge quality gates that block completion on failing smoke tests, and containerized the full stack for reproducible deployment across environments.

Credit Pulse | Full-Stack Python and Angular Application

2026

- Built a full-stack application in FastAPI and Angular 20, structured as three composable stages with explicit service contracts so each could be tested or replaced independently.
- Integrated third-party APIs (SEC EDGAR, Finnhub), normalizing inconsistent payloads and handling rate limits, missing fields, and upstream failures with graceful degradation.
- Implemented Server-Sent Events streaming so the client renders results progressively as each processing stage completes.
- Prototyped a server-driven UI approach in which the backend emits component specifications the client renders, with local model inference removing external service dependencies.

Freetown Village | User-Facing Web Application, Concept to Production

2024 - 2025

- Took a user-facing application from initial concept to deployed production, running stakeholder interviews, prototyping across three fidelity stages in Figma and Figma make, and shipping the React build on Netlify.
- Owned the application architecture end to end, including component structure, information architecture, and content structure, and defined a costed delivery roadmap for stakeholders.
- Iterated the build on usability testing findings, refining navigation and layout to reduce friction before public release.

EDUCATION

M.S., Human-Computer Interaction Indiana University Indianapolis GPA 3.8	2026
B.E., Computer Science Visvesvaraya Technological University 7.33/10 CGPA	2020

CERTIFICATIONS

Claude 101 and Claude Code 101, Anthropic	Jun 2026
Introduction to Git and GitHub, Coursera	Jun 2020

PUBLICATIONS

"You Need to Know What Not To Do": Co-Designing VR to Support Empathy and Social Decision-Making in Children. Interaction Design and Children (IDC), ACM, 2026. Co-author.

Recognition of Spontaneous Emotions in Virtual Reality Using Consumer Headset-Based Facial Tracking. MeaningfulXR 2026. First author, accepted.